

Bell-Scale Growth of OEIS A088714

A rigorous n th-root asymptotic, quantitative bounds,
and a separately identified finer conjecture

Part II: golden-ratio moment laws for A088714 and A088713

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Abstract

Let $A(x) = \sum_{n \geq 0} a_n x^n$ be the unique formal power series satisfying $A(x) = 1 + xA(x)^2A(xA(x))$. Its coefficients form OEIS A088714. We prove

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right), \quad a_n^{1/n} \sim \frac{n}{e \log n}.$$

The proof is based on an exact triangular recurrence obtained from composition iterates. A simple path through that recurrence supplies the lower bound. The upper bound comes from a coefficientwise inequality for shifted Poisson moments, proved here by a positivity argument involving Stirling numbers. More explicitly, for every $n \geq 1$,

$$\max_{1 \leq h \leq n+1} h^{n+1-h} \leq a_n \leq \inf_{\lambda > 0} \left(1 + \frac{2}{\lambda}\right)^n T'_n(\lambda),$$

where T_n is the n th Touchard polynomial. Consequently the ordinary generating function has radius zero, whereas the exponential generating function is entire. Exact coefficient computations accompany the proof. They suggest a considerably sharper Bell-number comparison, but that comparison and its multiplicative constant are not claimed as established results.

Part II (added October 1, 2026) realizes A088714 and its companion A088713 as the moment sequences of unique probability measures μ, ν on $[0, \infty)$, proves strict positivity of all Hankel minors, exact Stieltjes-transform equations, golden-ratio endpoint laws $F(s) \sim s^{-(3-\sqrt{5})/2}$ and $G(s) \sim s^{-(\sqrt{5}-1)/2}$ with leading constant one, the ratio laws $a_{n+1}/a_n \sim c_{n+1}/c_n \sim n/\log n$, and logarithmic far-tail laws. It uses the theorem of Part I and does not settle the finer conjecture.

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Part I

Bell-scale growth

1 The problem and the precise result

The OEIS entry A088714 defines the sequence through the formal functional equation [1]

$$A(x) = 1 + xA(x)^2A(xA(x)), \quad A(0) = 1. \quad (1.1)$$

Its beginning is

$$1, 1, 3, 13, 69, 419, 2809, 20353, 157199, 1281993, 10963825, \dots$$

The index convention throughout is $a_0 = 1$, $a_1 = 1$, $a_2 = 3$. All logarithms are natural. Formal composition is legitimate because the series substituted into A has zero constant term.

The composition in (1.1) makes this a different problem from ordinary algebraic generating-function asymptotics. In fact, the answer rules out a positive radius of convergence. One must not assume that the formal identity initially defines a holomorphic function near the origin.

Theorem 1.1 (Main asymptotic). *For the coefficients of (1.1),*

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (1.2)$$

Equivalently, at the level of n th-root asymptotics,

$$a_n^{1/n} = \frac{n}{e \log n} \left(1 + O\left(\frac{\log \log n}{\log n}\right)\right), \quad \lim_{n \rightarrow \infty} \frac{e \log n}{n} a_n^{1/n} = 1. \quad (1.3)$$

In particular,

$$a_n = \left(\frac{n}{e \log n}\right)^n \exp\left(O\left(\frac{n \log \log n}{\log n}\right)\right). \quad (1.4)$$

The $-n$ in (1.2) is important: it identifies the constant $1/e$ in (1.3). The weaker assertion $\log a_n \sim n \log n$ would miss both the logarithmic denominator and this constant.

What the theorem does not say. Equation (1.4) is not the multiplicative equivalent $a_n \sim (n/(e \log n))^n$. The error in its logarithm is $o(n)$, but it need not tend to zero. Likewise, an n th-root equivalent does not by itself imply an equivalent for a_n/a_{n-1} . (2026-10-01: Part II proves $a_{n+1}/a_n \sim n/\log n$ through the Stieltjes-moment representation and log-convexity, Theorem 12.3.) Section 8 discusses a finer numerical pattern without confusing it with the theorem.

A useful feature of the argument is that it gives bounds for every index, not merely an eventual limiting statement.

Theorem 1.2 (Finite-index comparison). *Define the Touchard polynomials by*

$$T_n(\lambda) = \sum_{j=0}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \lambda^j, \quad (1.5)$$

where $\left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\}$ denotes a Stirling number of the second kind, and let $T'_n(\lambda)$ mean differentiation with respect to λ . For every integer $n \geq 1$,

$$\max_{h \in \{1, \dots, n+1\}} h^{n+1-h} \leq a_n \leq \inf_{\lambda > 0} \left(1 + \frac{2}{\lambda}\right)^n T'_n(\lambda). \quad (1.6)$$

For every $r > 0$ and $\lambda > 0$ this implies the entirely explicit bound

$$a_n \leq \left(1 + \frac{2}{\lambda}\right)^n \frac{n!}{r^n} (e^r - 1) \exp(\lambda(e^r - 1)). \quad (1.7)$$

The remainder of the proof is self-contained. The OEIS supplies the sequence identification and related formal identities. Standard Bell number and Lambert W information is used for interpretation and numerical comparison, not to assume the answer for A088714.

2 Formal identities and the exact coefficient recurrence

2.1 Existence, uniqueness, and direct extraction

In (1.1), the coefficient of x^n on the right depends only on $a_0, \dots, a_{n-1}$. Starting with $a_0 = 1$ therefore determines the coefficients successively and proves existence and uniqueness in $\mathbb{Z}[[x]]$. The same observation proves their nonnegativity. Expanding the outer occurrence of A gives the exact identity

$$a_n = \sum_{k=0}^{n-1} a_k [x^{n-1-k}] A(x)^{k+2}, \quad n \geq 1. \quad (2.1)$$

This recurrence is useful for independently checking computed coefficients. It also explains the finer heuristic in Section 8, but it is not the most convenient recurrence for our rigorous upper bound.

2.2 Passage to composition iterates

Put $B(x) = xA(x)$. Equation (1.1) becomes

$$B(x) = x + B(x)B(B(x)). \quad (2.2)$$

Since $B(x) = x + O(x^2)$, it has a formal compositional inverse. Substitution of that inverse into (2.2) yields

$$B^{(-1)}(x) = x - xB(x) = x - x^2A(x). \quad (2.3)$$

This is the reversion identity also recorded in the OEIS entry [1].

For each integer $k \geq 0$, define

$$F_k(x) = \frac{B^{\circ k}(x)}{x}, \quad F_0(x) = 1, \quad F_1(x) = A(x). \quad (2.4)$$

Here $B^{\circ 0}(x) = x$ and $B^{\circ k}$ denotes k -fold composition, not an ordinary power. Composing (2.2) with $B^{\circ(k-1)}$ and dividing by x gives

$$F_k(x) = F_{k-1}(x) + xF_k(x)F_{k+1}(x), \quad k \geq 1. \quad (2.5)$$

The first few equations are the infinite system displayed in the OEIS entry; the derivation here fixes the indexing and proves the general identity [1].

Write

$$F_k(x) = \sum_{n \geq 0} c_{n,k} x^n.$$

Then

$$c_{0,k} = 1 \quad (k \geq 0), \quad c_{n,0} = 0 \quad (n \geq 1), \quad a_n = c_{n,1}. \quad (2.6)$$

Coefficient extraction in (2.5) gives

$$c_{n,k} - c_{n,k-1} = \sum_{i=0}^{n-1} c_{i,k} c_{n-1-i,k+1}. \quad (2.7)$$

Equivalently,

$$c_{n,k} = \sum_{j=1}^k \sum_{i=0}^{n-1} c_{i,j} c_{n-1-i,j+1}. \quad (2.8)$$

Every term on the right has first index less than n . Consequently this is a genuine triangular recurrence despite the occurrence of $k+1$. It proves positivity and monotonicity in k , and is the main structural tool of the article.

For illustration, the first three coefficient polynomials are

$$\begin{aligned} c_{1,k} &= k, \\ c_{2,k} &= k^2 + 2k, \\ c_{3,k} &= k^3 + 5k^2 + 7k. \end{aligned}$$

Evaluating at $k=1$ gives 1, 3, 13 as required. These are not ordinary Touchard polynomials; the Poisson polynomials introduced later are comparison objects, not an exact formula for the sequence.

2.3 Relation to the signed reversion sequence

For completeness, set

$$g(x) = -B^{(-1)}(-x) = x + x^2 A(-x).$$

The inverse identity implies

$$g(g(x)) = (1+x)g(x).$$

For $n \geq 2$ its coefficient of x^n is $(-1)^{n-2} a_{n-2}$. Thus the signed series in A067145 has the same coefficient magnitudes, with this explicit index shift [2]. This identity is an additional verification route, not an analytic-convergence argument.

3 A lower bound from an increasing-index path

Lemma 3.1. *For every $n \geq 0$ and every integer $k \geq 1$,*

$$c_{n,k} \geq k^n. \quad (3.1)$$

Proof. The case $n = 0$ is immediate. Suppose the statement holds for smaller first indices. By (2.8),

$$\begin{aligned} c_{n,k} &\geq \sum_{j=1}^k \sum_{i=0}^{n-1} j^i (j+1)^{n-1-i} \\ &= \sum_{j=1}^k ((j+1)^n - j^n) = (k+1)^n - 1 \geq k^n. \end{aligned}$$

The inner sum is a difference of two n th powers, and the outer sum then telescopes. This completes the induction. $\square$

The $i = 0$ summand of (2.7) and positivity imply

$$c_{n,k} \geq c_{n-1,k+1} \quad (n \geq 1).$$

Iterating this inequality $h - 1$ times, for $1 \leq h \leq n + 1$, gives

$$a_n = c_{n,1} \geq c_{n-h+1,h} \geq h^{n-h+1}. \quad (3.2)$$

This proves the lower half of Theorem 1.2.

3.1 Optimizing the lower bound

Let $N = n + 1$ and temporarily allow h to be real. The function

$$\phi_N(h) = (N - h) \log h$$

has derivatives

$$\phi'_N(h) = \frac{N}{h} - 1 - \log h, \quad \phi''_N(h) = -\frac{N}{h^2} - \frac{1}{h} < 0.$$

Its unique maximum is attained at

$$h_*(1 + \log h_*) = N, \quad h_* = \frac{N}{W(eN)}. \quad (3.3)$$

The positive real branch of Lambert W , characterized by $W(z)e^{W(z)} = z$, is intended [4]. The exact maximizing integer is among $\lfloor h_* \rfloor$ and $\lceil h_* \rceil$. If $w_* = W(eN)$, the continuous maximum is

$$\phi_N(h_*) = N \left(w_* - 2 + \frac{1}{w_*} \right). \quad (3.4)$$

Rounding to a nearest integer changes this value by $O((\log N)^2/N)$: on an interval of fixed length around h_* , $|\phi''_N(h)| = O((\log N)^2/N)$, and $\phi'_N(h_*) = 0$. This provides an accurate, explicit lower bound even when the leading logarithmic expression is numerically crude.

For the main theorem, a simpler choice suffices. Put

$$L = \log n, \quad \ell = \log \log n, \quad h = \left\lfloor \frac{n}{L} \right\rfloor.$$

As $n \rightarrow \infty$,

$$\log h = L - \ell + O(L/n), \quad h = n/L + O(1).$$

Hence (3.2) yields

$$\log a_n \geq n(L - \ell - 1) + \frac{n\ell}{L} - O(L). \quad (3.5)$$

In particular the lower bound has the same $n \log n$, $-n \log \log n$, and $-n$ terms as the claimed answer.

4 The Poisson-moment comparison inequality

The upper bound requires a comparison family that respects the increase from k to $k + 1$ in (2.7). Ordinary exponential bounds are too small, while a rough factorial bound loses the logarithmic factor. Shifted Poisson moments provide the needed intermediate scale.

Let X have the Poisson distribution with mean $\lambda > 0$, and define

$$M_n(k) = \mathbb{E}(k + X)^n, \quad T_n(\lambda) = M_n(0). \quad (4.1)$$

Their exponential generating functions are

$$\sum_{n \geq 0} M_n(k) \frac{z^n}{n!} = \exp(kz + \lambda(e^z - 1)), \quad \sum_{n \geq 0} T_n(\lambda) \frac{z^n}{n!} = \exp(\lambda(e^z - 1)). \quad (4.2)$$

Expanding the last exponential proves (1.5); at $\lambda = 1$ these polynomials give the Bell numbers. Standard Bell identities are recorded in [3]; the identities needed below also follow directly from (4.2).

For $n \geq 1$ set

$$D_n(k) = M_n(k) - M_n(k - 1), \quad S_n(k) = \sum_{i=0}^{n-1} M_i(k) M_{n-1-i}(k + 1). \quad (4.3)$$

The central estimate is the following.

Lemma 4.1 (Shifted Poisson convolution inequality). *For every $n \geq 1$, $k \geq 1$, and $\lambda > 0$,*

$$S_n(k) \leq \left(1 + \frac{2}{\lambda}\right) D_n(k). \quad (4.4)$$

In fact, after multiplication by λ , the difference between the right and left sides is a polynomial with nonnegative coefficients in λ and $k - 1$.

4.1 Reduction to the shift $k = 1$

Write $u = k - 1$. The binomial theorem gives

$$D_n(k) = \sum_{r=1}^n \binom{n}{r} u^{n-r} D_r(1). \quad (4.5)$$

There is a matching identity for the ordinary convolution:

$$S_n(k) = \sum_{r=1}^n \binom{n}{r} u^{n-r} S_r(1). \quad (4.6)$$

To verify it, expand both shifted moments in the definition of $S_n(k)$. For fixed nonnegative integers p, q with $p + q = r - 1$, the coefficient sum over the convolution index is

$$\sum_i \binom{i}{p} \binom{n-1-i}{q} = \binom{n}{p+q+1} = \binom{n}{r}.$$

This identity counts a subset of r elements of an n -element ordered set by the location of its $(p+1)$ st element. Thus it holds including all boundary cases. Equations (4.5) and (4.6) reduce the lemma to $k = 1$.

Differentiating (4.2) with respect to λ shows

$$D_n(1) = T'_n(\lambda). \quad (4.7)$$

Suppress the argument λ temporarily, write $S_n = S_n(1)$, and set

$$H_n = (\lambda + 2)T'_n - \lambda S_n. \quad (4.8)$$

We shall prove that H_n has nonnegative coefficients for every $n \geq 1$.

4.2 A recurrence for the difference polynomial

The moment identities imply

$$T_{n+1} = \lambda(T_n + T'_n), \quad (4.9)$$

$$M_{n+1}(k) = (k + \lambda)M_n(k) + \lambda M'_n(k). \quad (4.10)$$

Here and below a prime means differentiation in λ , not in k . Applying (4.10) to both factors in S_n gives

$$S_{n+1} = \frac{\lambda}{2}S'_n + \left(\lambda + \frac{3}{2}\right)S_n + \frac{1}{2}(M_n(1) + M_n(2)). \quad (4.11)$$

Indeed, the two shifted convolution sums add up to $2S_{n+1} - M_n(1) - M_n(2)$. Furthermore,

$$M_n(1) = T_n + T'_n, \quad M_n(2) = T_n + 2T'_n + T''_n.$$

Substitute these formulas and (4.9) into (4.8). A direct simplification gives the exact recurrence

$$H_{n+1} = \frac{\lambda}{2}H'_n + (\lambda + 1)H_n + K_n, \quad (4.12)$$

where

$$K_n = \frac{\lambda(\lambda + 1)}{2}T''_n - 2\lambda T'_n + 2T_n. \quad (4.13)$$

The initial difference polynomials are

$$\begin{aligned} H_1 &= 2, \\ H_2 &= 2 + 2\lambda, \\ H_3 &= 2 + 6\lambda + \lambda^2, \\ H_4 &= 2 + 14\lambda + 8\lambda^2. \end{aligned}$$

They all have nonnegative coefficients.

4.3 Why the forcing polynomial is nonnegative

From (1.5), the coefficient of λ^j in K_n is

$$[\lambda^j]K_n = \frac{(j-1)(j-4)}{2} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} + \frac{j(j+1)}{2} \left\{ \begin{matrix} n \\ j+1 \end{matrix} \right\}. \quad (4.14)$$

For $n \geq 4$, this is visibly nonnegative except possibly when $j = 2$ or $j = 3$. Those cases require only

$$\left\{ \begin{matrix} n \\ 2 \end{matrix} \right\} \leq 3 \left\{ \begin{matrix} n \\ 3 \end{matrix} \right\} \quad (n \geq 3), \quad \left\{ \begin{matrix} n \\ 3 \end{matrix} \right\} \leq 6 \left\{ \begin{matrix} n \\ 4 \end{matrix} \right\} \quad (n \geq 4). \quad (4.15)$$

Here is a common elementary proof. If a partition of n elements into r blocks has $n > r$, it has a nonsingleton block. Choose such a block by any fixed deterministic rule and split off a specified element of it. Remember the unordered pair of blocks created. Merging that pair recovers the original partition. This is an injection into partitions with $r + 1$ blocks together with a choice of two of those blocks. Therefore

$$\left\{ \begin{matrix} n \\ r \end{matrix} \right\} \leq \binom{r+1}{2} \left\{ \begin{matrix} n \\ r+1 \end{matrix} \right\} \quad (n > r). \quad (4.16)$$

Taking $r = 2, 3$ proves (4.15). Thus K_n has nonnegative coefficients for all $n \geq 4$.

Starting from H_4 , recurrence (4.12) preserves coefficientwise nonnegativity: differentiation, multiplication by $\lambda/2$ or by $\lambda + 1$, and addition of K_n all preserve it. Together with the three earlier base cases, this proves $H_n \geq 0$ coefficientwise for all $n \geq 1$. By (4.8) and (4.7),

$$\lambda S_n(1) \leq (\lambda + 2)D_n(1).$$

Finally, (4.5)–(4.6) extend the result to all $k \geq 1$, proving Lemma 4.1.

5 The global upper bound and completion of the proof

5.1 Comparison with the triangular recurrence

Fix any $\lambda > 0$ and write

$$\gamma = 1 + \frac{2}{\lambda}.$$

We claim that, simultaneously for all $n \geq 0$ and $k \geq 0$,

$$c_{n,k} \leq \gamma^n M_n(k). \quad (5.1)$$

For $n = 0$, both sides equal 1. Assume the bound for smaller first indices. Equations (2.8) and (4.4) imply, for $n \geq 1$,

$$\begin{aligned} c_{n,k} &\leq \gamma^{n-1} \sum_{j=1}^k S_n(j) \\ &\leq \gamma^n \sum_{j=1}^k (M_n(j) - M_n(j-1)) \\ &= \gamma^n (M_n(k) - M_n(0)) \leq \gamma^n M_n(k). \end{aligned} \quad (5.2)$$

The boundary $k = 0$ is immediate. This proves the induction.

Importantly, the penultimate expression is sharper than the induction claim. At $k = 1$, (4.7) yields

$$a_n \leq \left(1 + \frac{2}{\lambda}\right)^n T'_n(\lambda). \quad (5.3)$$

Because λ was arbitrary, this proves the upper half of Theorem 1.2.

Differentiation of (4.2) gives

$$\sum_{n \geq 0} T'_n(\lambda) \frac{r^n}{n!} = (e^r - 1) \exp(\lambda(e^r - 1)).$$

All coefficients are nonnegative. Any one term is bounded by the whole sum when $r > 0$, and (1.7) follows. This argument is just a positive-coefficient bound; it does not require any analyticity of $A(x)$.

5.2 An explicit parameter choice

Let $L = \log n$ and $\ell = \log L$ as before. For all sufficiently large n , take

$$\lambda = L, \quad r = L - 2\ell > 0. \quad (5.4)$$

Then $e^r = n/L^2$. Taking logarithms in (1.7), and using $\log(e^r - 1) \leq r$, gives

$$\begin{aligned} \log a_n &\leq \log n! - n \log r + n \log(1 + 2/L) + L(e^r - 1) + r \\ &= \log n! - n \log(L - 2\ell) + n \log(1 + 2/L) + \frac{n}{L} - 2\ell. \end{aligned} \quad (5.5)$$

The elementary estimates

$$\begin{aligned} \log n! &= nL - n + O(L), \\ \log(L - 2\ell) &= \ell - \frac{2\ell}{L} + O\left(\frac{\ell^2}{L^2}\right), \\ \log(1 + 2/L) &= \frac{2}{L} + O(L^{-2}) \end{aligned}$$

now show

$$\log a_n \leq n(L - \ell - 1) + \frac{2n\ell + 3n}{L} + O\left(\frac{n\ell^2}{L^2} + L\right). \quad (5.6)$$

For the first estimate, even Stirling's full formula is unnecessary: bounding the sum $\sum_{j=1}^n \log j$ between its neighboring integrals already gives $\log n! = n \log n - n + O(\log n)$.

The lower estimate (3.5) and upper estimate (5.6) together give (1.2). Dividing by n and exponentiating proves (1.3) and (1.4). This completes the proof of Theorem 1.1.

Remark 5.1 (Why a tunable comparison is necessary). A bound of the form $a_n \leq C^n n!$ cannot identify the factor $(\log n)^{-n}$. Even a comparison with C^n times a fixed Bell-type sequence leaves an unwanted constant C in the n th root. Here the factor $1 + 2/\lambda$ can approach 1, and the valid choice $\lambda = \log n$ makes its total logarithmic cost only $O(n/\log n)$. The dependence of the Poisson mean on n is fully allowed because (5.3) holds for every positive λ at every index.

6 Consequences and comparison with Bell numbers

6.1 Superexponential but subfactorial growth

The main theorem immediately implies

$$\frac{a_n}{C^n} \longrightarrow \infty \quad \text{for every fixed } C > 0. \quad (6.1)$$

On the other hand,

$$\left(\frac{a_n}{n!}\right)^{1/n} = \frac{1}{\log n} \left(1 + O\left(\frac{\log \log n}{\log n}\right)\right) \longrightarrow 0. \quad (6.2)$$

Thus $a_n/(C^n n!) \rightarrow 0$ for every fixed $C > 0$. These two assertions locate the growth more precisely than the phrase “almost factorial”: there is a factor of approximately $(\log n)^n$ between $n!$ and a_n on the logarithmic scale.

For every fixed $s < 1$ and $C > 0$,

$$\frac{a_n}{C^n (n!)^s} \longrightarrow \infty,$$

because the logarithm of this quotient has dominant term $(1-s)n \log n$. In the convention that Gevrey order s means a bound by $CK^n(n!)^s$, the series has exact Gevrey order 1, with zero factorial exponential type. This statement specifies the convention, as Gevrey indexing is not uniform across the literature.

6.2 The two generating functions behave differently

By the Cauchy–Hadamard formula,

$$\text{rad } A = \frac{1}{\limsup_{n \geq 0} a_n^{1/n}} = 0.$$

In particular, there is no holomorphic germ at zero whose Taylor series has these coefficients and satisfies (1.1). The formal series itself remains perfectly well-defined and unique.

In contrast, the exponential generating function

$$\mathcal{A}(t) = \sum_{n \geq 0} a_n \frac{t^n}{n!} \quad (6.3)$$

is entire by (6.2). Summing the finite-index upper bound also gives, for every $t \geq 0$ and $\lambda > 0$,

$$\mathcal{A}(t) \leq 1 + (e^{\gamma t} - 1) \exp(\lambda(e^{\gamma t} - 1)), \quad \gamma = 1 + 2/\lambda. \quad (6.4)$$

Corollary 6.1 (Double-exponential scale). *Along the positive real axis,*

$$\lim_{t \rightarrow \infty} \frac{\log \log \mathcal{A}(t)}{t} = 1. \quad (6.5)$$

Proof. For a fixed λ , (6.4) bounds the limit superior by $1 + 2/\lambda$. Letting $\lambda \rightarrow \infty$ gives the upper bound 1. For the lower bound fix $0 < \varepsilon < 1$ and take $n = \lfloor e^{(1-\varepsilon)t} \rfloor$. The single term $a_n t^n / n!$ and Theorem 1.1 give

$$\log \left(\frac{a_n t^n}{n!} \right) = n(\log t - \log \log n + o(1)) = n(-\log(1 - \varepsilon) + o(1)).$$

The constant inside the last parentheses is positive. Taking a further logarithm shows that the limit inferior in (6.5) is at least $1 - \varepsilon$. Let $\varepsilon \downarrow 0$. $\square$

Since the coefficients of $\mathcal{A}$ are nonnegative, its maximum modulus on $|z| = t$ equals $\mathcal{A}(t)$. Consequently its usual entire function order is infinite. Equation (6.5), rather than just “infinite order,” states the useful growth scale.

6.3 Bell-number scale is not Bell-number equivalence

Let $B_n = T_n(1)$ denote the Bell numbers. A standard asymptotic, with $w = W(n)$, is [3]

$$B_n \sim \frac{\exp(n(w - 1 + 1/w) - 1)}{\sqrt{1 + w}}. \quad (6.6)$$

In particular $B_n^{1/n} \sim n/(e \log n)$. Therefore our theorem implies

$$\left(\frac{a_n}{B_n} \right)^{1/n} \longrightarrow 1, \quad a_n = B_n \exp(o(n)). \quad (6.7)$$

These are the senses in which the sequence has Bell-number-scale growth. They do not assert that a_n/B_n is bounded or converges. Indeed, the finite computations in the next section show a substantial and growing discrepancy. The formula (6.6) is a comparison formula for Bell numbers, not a formula silently transferred to A088714.

7 Exact computation and numerical diagnostics

7.1 A triangular dynamic program

To obtain $a_0, \dots, a_N$, store $c_{0,k} = 1$ for $0 \leq k \leq N+1$. For each $n = 1, \dots, N$, compute

$$c_{n,0} = 0, \quad c_{n,k} = c_{n,k-1} + \sum_{i=0}^{n-1} c_{i,k} c_{n-1-i,k+1} \quad (1 \leq k \leq N-n+1).$$

Then output $a_n = c_{n,1}$. The triangular cutoff is sufficient: every requested entry on the right was computed at an earlier first index. The number of products is exactly

$$\sum_{n=1}^N n(N-n+1) = \frac{N(N+1)(N+2)}{6}.$$

Thus the algorithm uses $O(N^3)$ arbitrary-precision arithmetic operations and $O(N^2)$ integer cells. These are arithmetic counts, not unit-cost bit-complexity claims; the integers grow with N .

The archive contains both a dependency-free Python implementation and a faster C++ implementation using GMP. The supplied data were computed with exact integer arithmetic through $n = 1200$. The Python and C++ outputs agree through $n = 400$. Independent truncated-polynomial checks verify (1.1) and (2.3); the supplied verification script also checks the auxiliary polynomial recurrences and finite bounds on explicitly reported finite ranges.

7.2 What the diagnostics measure

Define

$$Q_n = \frac{e \log n}{n} a_n^{1/n}, \quad (7.1)$$

$$R_n = \frac{a_n}{a_{n-1}}, \quad (7.2)$$

$$D_n^{\text{num}} = \log \left(\frac{a_n}{B_n} \right) - W(n)^2 - 3W(n). \quad (7.3)$$

The notation D_n^{num} is distinct from the Poisson moment difference $D_n(k)$ in Section 4. The main theorem proves $Q_n \rightarrow 1$. The quantities R_n and D_n^{num} examine finer behavior not established by that theorem. (2026-10-01: the first-order law $R_n \sim n/\log n$ is now proved in Part II, Theorem 12.3; D_n^{num} and Conjecture 8.1 remain open.)

n	$\log_{10} a_n$	Q_n	$R_n - n/W(n)$	D_n^{num}
20	17.180354	2.942966	2.650843	-3.494768
50	53.134060	2.457011	2.312570	-3.262262
100	123.688846	2.159875	2.205968	-3.172516
200	286.269488	1.944240	2.146342	-3.115335
400	657.374892	1.791450	2.110948	-3.078375
800	1495.731414	1.682461	2.089104	-3.054662
1200	2408.704723	1.633120	2.080197	-3.045054

Table 1: Diagnostics from independently generated exact coefficients. Only the limit $Q_n \rightarrow 1$ is proved here. The final column tests the conjecture of Section 8.

The relatively slow approach of Q_n to 1 is not surprising: $\log \log n / \log n$ is still appreciable at these indices. Substituting only the first term $(n/(e \log n))^n$ is therefore not a numerically accurate approximation to a_n , despite its correct n th-root scale. The optimized finite lower and upper bounds offer certified alternatives when actual error control is required.

The normalization in (7.3) is motivated next. Its numerical behavior is evidence for a conjecture, not evidence that upgrades Theorem 1.1 to a multiplicative equivalent. In particular no rigorous enclosure for a limiting constant is extracted from the table.

8 A finer asymptotic: formal motivation and its missing step

8.1 The leading endpoint balance

Write $R_n = a_n/a_{n-1}$ and consider (2.1). Near its outer endpoint put $k = n - 1 - m$. Then the corresponding summand is

$$a_{n-1-m}[x^m]A(x)^{n+1-m}.$$

Since $a_1 = 1$, the first approximation to the coefficient is $n^m/m!$. If neighboring ratios can be treated as locally constant, the endpoint sum divided by a_n is approximately

$$\frac{1}{R_n} \sum_{m \geq 0} \frac{(n/R_n)^m}{m!} = \frac{e^{n/R_n}}{R_n}.$$

Setting this expression equal to 1 leads to

$$R_n \log R_n \approx n, \quad R_n \approx \frac{n}{W(n)}.$$

This explains the natural Lambert W scale of the ratios. It is not a proof: local ratio regularity and uniform control of the other summands have been assumed in deriving the approximation.

8.2 The first correction and a candidate normalization

There is more information in the same calculation. From the first few coefficients,

$$\log A(x) = x + \frac{5}{2}x^2 + O(x^3).$$

Expanding the near-endpoint coefficient for m of logarithmic size gives the formal local approximation

$$[x^m]A(x)^{n+1-m} = \frac{n^m}{m!} \left(1 + \frac{3m(m-1)}{2n} + \dots \right). \quad (8.1)$$

The factor $3/2$ combines the quadratic term of $\log A$ with the replacement of $n + 1 - m$ by n .

Suppose, for this calculation, that the ratio has a smooth expansion near $R_n = n/w$, where $w = W(n)$. Put

$$t = n/R_n, \quad d = \frac{w}{1+w}.$$

The local variation of the ratios contributes $m(m+1)d/(2n)$ to the bracket multiplying the Poisson kernel. Averaging $m(m-1)$ and $m(m+1)$ against that kernel replaces them by t^2 and $t^2 + 2t$, respectively. The opposite endpoint, where one inner factor has almost all the degree, begins with $2a_{n-1}$ from $A(x)^2$. At this formal order the equation becomes

$$1 \approx \frac{e^t}{R_n} \left(1 + \frac{(3+d)t^2 + 2dt}{2n} \right) + \frac{2}{R_n}. \quad (8.2)$$

Writing $R_n = n/w + \delta$ and expanding gives

$$\delta(1+w) \approx 2 + \frac{(3+d)w + 2d}{2},$$

and hence the candidate ratio approximation

$$R_n \approx \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1+W(n))^2}. \quad (8.3)$$

This is consistent with the exact data, but the approximation signs in this subsection are intentional.

A normalization whose successive quotient has these same first terms is

$$\mathcal{N}_n = \frac{\exp(n(w-1+1/w) - 1 + w^2 + 3w)}{\sqrt{1+w}}, \quad w = W(n). \quad (8.4)$$

Indeed, using $w' = w/(n(1+w))$ and Taylor expansion in the real variable n shows

$$\frac{\mathcal{N}_n}{\mathcal{N}_{n-1}} = \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1+W(n))^2} + O\left(\frac{W(n)^3}{n}\right). \quad (8.5)$$

One way to check the differentiation is to note that

$$\frac{d}{dn} [n(w-1+1/w)] = w.$$

The additional derivative from $w^2 + 3w - \frac{1}{2} \log(1+w)$ supplies the constant and rational correction in (8.5).

Conjecture 8.1 (Finer Bell comparison). *There exists a constant $C > 0$ such that, with $w = W(n)$,*

$$a_n \sim C B_n \exp(w^2 + 3w) \sim C \mathcal{N}_n. \quad (8.6)$$

The existence or numerical value of C is not established in this article. The data and the endpoint calculation motivate the displayed form, rather than merely an arbitrary fit to an exponential correction.

8.3 A precise sufficient condition for the conjecture

The unresolved analytic step can be stated cleanly.

Proposition 8.2 (Conditional implication). *Suppose that for some fixed $p \geq 0$ the actual ratios satisfy*

$$\frac{a_n}{a_{n-1}} = \frac{n}{W(n)} + 2 + \frac{W(n)}{2(1+W(n))^2} + O\left(\frac{W(n)^p}{n}\right). \quad (8.7)$$

Then Conjecture 8.1 follows with some finite, strictly positive constant C .

Proof. Compare (8.7) with (8.5). Since both ratios are asymptotic to $n/W(n)$, their quotient is

$$1 + O\left(\frac{W(n)^{\max(p,3)+1}}{n^2}\right).$$

The error is summable because $W(n) = O(\log n)$. Therefore the successive differences of $\log(a_n/\mathcal{N}_n)$ are absolutely summable. This logarithm converges to a finite real number, so $a_n/\mathcal{N}_n$ tends to a finite positive constant. Formula (6.6) then gives the Bell-number version. $\square$

The main theorem does not supply (8.7). In particular, it does not justify replacing all nearby coefficient ratios by a smooth function, nor does it control every non-endpoint contribution at a summable error level. A proof of such uniform endpoint estimates would be a genuine strengthening of the present result. This identifies the remaining task without presenting a formal asymptotic calculation as a completed proof.

9 Reproducibility and verification scope

The archive is organized so that the mathematical theorem does not depend on a numerical black box. The principal files are `article.tex`, `article.pdf`, the exact data file `data/coefficients.txt`, and the scripts under `code/`.

The dependency-free command

```
python3 code/verify.py
```

checks the initial OEIS terms; verifies the defining composition and inverse identities by independent truncated-polynomial arithmetic; checks the Poisson difference-polynomial identity and coefficient signs on a finite range; and tests both sides of the finite-index bound using exact integer arithmetic. Its actual output is supplied in `data/verification.txt`. These are useful implementation audits; the all-index positivity proof is the argument in Section 4, not an extrapolation from those finite checks.

The command

```
python3 code/coefficients.py -n 400 -output coefficients.txt
```

recomputes the first 401 terms without third-party Python packages. For a larger table, `code/compute_gmp.cpp` provides the same recurrence in C++ with GMP. Instructions in `README.md` describe compilation and analysis. The diagnostic calculations use exact integers as input and arbitrary-precision floating-point logarithms only at the presentation stage; no computed coefficient is rounded.

No computer-assisted proof of the finer conjecture is claimed. The article's rigorous conclusion is Theorem 1.1, supported by the explicit all-index inequalities of Theorem 1.2.

10 Conclusion

OEIS A088714 has the precise n th-root growth

$$a_n^{1/n} \sim \frac{n}{e \log n}.$$

The accompanying logarithmic error estimate is

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right).$$

The mechanism is visible in the proof. Composition iterates turn the functional equation into a positive triangular recurrence. Moving along its increasing-index direction gives roughly h^{n-h} , whose optimum occurs at h of order $n/\log n$. Shifted Poisson moments furnish a compatible upper comparison and recover the same leading scale, including the factor $1/e$.

The result determines the growth at logarithmic precision $o(n)$ and settles the convergence classification of both generating functions. The finer proposed factor $\exp(W(n)^2 + 3W(n))$ remains explicitly separate from the proved theorem.

Part II

Golden-ratio moment laws for A088714 and A088713

11 Source, scope and notation of Part II

Summary. Let $A(z) = \sum a_n z^n$ be the formal solution of $A(z) = 1 + zA(z)^2 A(zA(z))$, and let $C(z) = (1 - zA(z))^{-1} = \sum c_n z^n$. These are OEIS A088714 and A088713. Part II constructs unique probability measures μ, ν on $[0, \infty)$ with moments a_n, c_n . The construction factors a nonlinear series transformation into two positivity-preserving operator constructions. It proves strict total positivity of both Hankel matrices, and strict log-convexity after any finite number of iterations of the log-convexity operator. Their Stieltjes transforms $F(s)$ and $G(s)$ satisfy exact identities $F(s) = 1 - sF(s)^2 F(sF(s))$ and $G(s) = (1 + sF(s))^{-1}$ for $s > 0$. Writing $\alpha = (3 - \sqrt{5})/2$ and $\beta = (\sqrt{5} - 1)/2$, Part II proves $F(s) \sim s^{-\alpha}$ and $G(s) \sim s^{-\beta}$, including the leading constants. Consequently the distribution functions at zero are asymptotic to $\sin(\pi\alpha)t^\alpha/(\pi\alpha)$ and $\sin(\pi\beta)t^\beta/(\pi\beta)$, respectively. Compact algebraic approximants bracket F monotonically; their exponents are ratios of Fibonacci numbers. Combining the moment structure with the Bell-scale growth bound of Part I (Theorem 1.1) yields $a_{n+1}/a_n \sim c_{n+1}/c_n \sim n/\log n$ and $\log \mu([x, \infty)) \sim \log \nu([x, \infty)) \sim -x \log x$. The package includes independent exact checks, outward-rounded orbit enclosures, and a research agenda. These are written mathematical proofs, not Lean-checked theorems; historical priority is not asserted as certified.

Provenance. Part II is manuscript 35 of batch 73 of the repository’s incoming-reports intake (cluster O2): *Golden-ratio moment laws for OEIS A088714 and A088713. Positive operators, strict Hankel positivity, ratio asymptotics, and two endpoint laws*, by OpenAI ChatGPT, prepared for Vladimir Reshetnikov and dated October 1, 2026 (a 1,524-line source and a 23-page PDF). It arrived as `golden_moment_laws.zip` in commit `f8c3a392a` and was placed in commit `6e193dd4f`. The manuscript pins the repository at commit `1f1981f682b2878bde51a6ad40c22777f362fc05` and identifies Part I by the Git blob `ccb50ef335ac4de27bd98a5a152fb48c978c7a96` of its source; that blob was still the source of this report at the placement commit, so the manuscript read the text now printed as Part I. Its code, data and source audit are shipped with the prefix `02-golden-moments-`; its own L^AT_EX source, README and PDF are not shipped and survive in the archive of the arrival commit. Part I was written independently, on September 20, 2026; the two Parts share no text.

What is not reprinted. Part I proves the Bell-scale growth theorem (Theorem 1.1, with the finite bounds of Theorem 1.2); the golden-moment manuscript re-proves it in an appendix that is not reprinted here. That appendix runs through Part I’s Sections 2–5 step for step: the triangular recurrence (2.8), the lower bound of Lemma 3.1, the Poisson comparison of Lemma 4.1 with the same polynomials H_n, K_n and the same initial values, the merge inequality (4.16), and the parameter choice (5.4). It states nothing that Part I does not, and the manuscript itself calls it “not claimed as new work”. The manuscript included it so that its deductions would not depend on accepting the earlier report; inside one report that purpose lapses, and every use of the growth bound below refers to Part I. Everything else in the manuscript is printed: all sections, all proofs, remarks and research questions, and its dependency and novelty audit (Appendix A).

Editorial changes. Section k of the manuscript is Section $k + 11$ here, and its Appendix B is Appendix A; every label carries the prefix `gml:`. References to “the earlier ProveIt article” now point to Part I, and the manuscript’s link to a GitHub copy of Part I at the pin is replaced by an internal reference. The table fragments and file names in Section 22 are the shipped `02-golden-moments-`files. The manuscript’s bibliography entry for A088714 is merged with Part I’s. Nothing else in the mathematics was changed. Two dated notes in Part I (Section 8 and the paragraph after Theorem 1.1) record what Part II settles; Conjecture 8.1 is not settled by it (Section 23).

Status. Part II, like Part I, is AI-assisted and unrefereed, and it is not formalized: no Lean or Rocq development checks any statement of this report, and placement in this collection confers no formal status. Its novelty audit is bounded (Section 12.3 and Appendix A).

Related repository material. The positive inverse $U^{-1}(t) = t(1 + U(t))$ of Theorem 12.1 and the determinacy argument through an entire exponential generating function belong to the circle of ideas of the transseries package *Moment Determinacy and Nonlinear Transseries* in the directory

```
Analysis/Transseries/docs/series-and-transseries/
Moment_Determinacy_Nonlinear_Transseries/
```

which treats general positive Stieltjes inverses $T(y) = yS(T(y))$. That package does not mention A088714 or A088713, and Part II does not depend on it.

Notation. The two Parts were written independently and reuse letters. Every symbol of Part II is local to Part II; Table 2 lists each letter that also occurs in Part I with both meanings. No symbol of the manuscript was renamed.

12 The sequences, the contribution, and its scope

12.1 Formal definitions and the earlier result

Throughout, $a_0 = c_0 = 1$, logarithms are natural, and a power series without an explicitly specified analytic domain is a *formal* power series. The defining equations are

$$A(z) = 1 + zA(z)^2A(zA(z)), \quad (12.1)$$

$$C(z) = \frac{1}{1 - zA(z)}, \quad A(z) = C(zA(z)). \quad (12.2)$$

The second identity in (12.2) follows from (12.1); it is also recorded in the OEIS entries [1, 5]. Coefficients are determined successively because the right side of (12.1) at degree $n \geq 1$ uses only coefficients of A of degrees below n . Thus there is exactly one solution with constant term 1; its coefficients are positive integers. The initial values are

n	0	1	2	3	4	5	6	7
a_n	1	1	3	13	69	419	2809	20353
c_n	1	1	2	6	24	118	674	4308.

The notation C is used here for the generating function called A in OEIS A088713, whose defining identity is $C(z/C(z)) = (1 - z)^{-1}$. For verification, put $B = zA$. Equation (12.1) gives $B^{(-1)}(z) = z - z^2A(z) = z/C(z)$. Composing $A = C \circ B$ with this inverse and substituting in (12.2) proves the identity for C .

Table 2: Letters used in both Parts.

letter	Part I	Part II
c, C	$c_{n,k} = [x^n]F_k(x)$, coefficients of the composition iterates (2.4); C is the constant of Conjecture 8.1	c_n is A088713 and $C(z) = \sum c_n z^n = (1 - zA(z))^{-1}$
F, G	$F_k(x) = B^{\circ k}(x)/x$, formal series; $F_1 = A$	$F(s), G(s), F_r(s), G_r(s)$ are Stieltjes transforms, functions of $s > 0$; $F(s)$ has the asymptotic expansion $\sum (-1)^j a_j s^j$ as $s \downarrow 0$. Part II's $F_1(s) = 2/(1 + \sqrt{1 + 4s})$ is <i>not</i> Part I's $F_1(x) = A(x)$
B	$B(x) = xA(x)$	$B(z)$ is a moment series (Lemma 13.1); the iterates B_r of (14.1) approximate A , not xA
T	$T_n(\lambda)$, Touchard polynomials	$\mathcal{T}$ is the nonlinear transform (13.6); T is a multiplication operator inside proofs
D	$D_n(k)$, Poisson differences (4.3); D_n^{num} (7.3)	$D_n^{(h)}(m)$, Hankel determinants (Proposition 15.4)
H, K, S, M	polynomials H_n, K_n in λ ; shifted Poisson moments $M_n(k)$ and convolutions $S_n(k)$	$H(s)$ and K in Lemma 19.1; kernels H_M, S_M ; moment series $M(z)$
R, r	$R_n = a_n/a_{n-1}$ (7.2)	$R(z)$ is the renewal series (13.1); $r_n = m_{n+1}/m_n$ is a <i>forward</i> ratio, so Part I's R_n is r_{n-1} for $m = a$
Q, X	Q_n (7.1); X a Poisson variable	Q and X are positive operators (Lemmas 13.1 and 13.2)
γ, L, r	$\gamma = 1 + 2/\lambda$; $L = \log n$ and $r = L - 2\ell$ (5.4)	γ is a Tauberian exponent; $\mathcal{L}$ is the log-convexity operator; r indexes the iterates B_r, F_r

Part I of this report, *Bell-Scale Growth of OEIS A088714*, proves (Theorem 1.1)

$$\log a_n = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (12.3)$$

Part I explicitly separates this result from consecutive-coefficient asymptotics and from a finer multiplicative Bell-number conjecture. The golden-moment manuscript re-proves this bound, with the finite bounds of Theorem 1.2, in an appendix that is not reprinted here (Section 11). Part I's proof uses no numerical evidence, so the deductions below rest on written proofs only.

12.2 Main results

A normalized Stieltjes moment sequence is a sequence $m_n = \int_{[0,\infty)} t^n d\sigma(t)$ for a probability measure σ with all moments finite. It is *determinate* when this measure is unique. Define

$$\alpha = \frac{3 - \sqrt{5}}{2}, \quad \beta = \frac{\sqrt{5} - 1}{2}, \quad \alpha + \beta = 1, \quad \beta^2 + \beta = 1. \quad (12.4)$$

Theorem 12.1 (Moment and analytic structure). *There are unique probability measures μ, ν on $[0, \infty)$ such that*

$$a_n = \int t^n d\mu(t), \quad c_n = \int t^n d\nu(t) \quad (n \geq 0). \quad (12.5)$$

Both have all positive exponential moments finite. Their supports are unbounded, and all minors of their infinite Hankel matrices with strictly increasing row and column index lists are strictly positive.

For $(\mathcal{L}m)_n = m_n m_{n+2} - m_{n+1}^2$, every $\mathcal{L}^k a$ and $\mathcal{L}^k c$, $k \geq 1$, is a Stieltjes moment sequence with strictly positive entries.

For $s > 0$, set

$$F(s) = \int \frac{d\mu(t)}{1+st}, \quad G(s) = \int \frac{d\nu(t)}{1+st}. \quad (12.6)$$

These transforms satisfy

$$F(s) = 1 - sF(s)^2 F(sF(s)), \quad G(s) = \frac{1}{1 + sF(s)}. \quad (12.7)$$

Moreover, $U(s) = sF(s)$ is an increasing homeomorphism of $(0, \infty)$ onto itself and has inverse

$$U^{-1}(t) = t(1 + U(t)). \quad (12.8)$$

Theorem 12.2 (Golden-ratio endpoint laws). *As $s \rightarrow \infty$,*

$$F(s) \sim s^{-\alpha}, \quad G(s) \sim s^{-\beta}. \quad (12.9)$$

As $t \downarrow 0$,

$$\mu([0, t]) \sim \frac{\sin(\pi\alpha)}{\pi\alpha} t^\alpha, \quad \nu([0, t]) \sim \frac{\sin(\pi\beta)}{\pi\beta} t^\beta. \quad (12.10)$$

Neither measure has an atom at zero. For $q > 0$, its negative q th moment is finite precisely when $q < \alpha$ for μ , or $q < \beta$ for ν .

Theorem 12.3 (Consecutive ratios and the opposite tail). *The two sequences satisfy*

$$\frac{a_{n+1}}{a_n} \sim \frac{n}{\log n}, \quad \frac{c_{n+1}}{c_n} \sim \frac{n}{\log n}. \quad (12.11)$$

Their representing measures satisfy

$$\lim_{x \rightarrow \infty} \frac{\log \mu([x, \infty))}{x \log x} = \lim_{x \rightarrow \infty} \frac{\log \nu([x, \infty))}{x \log x} = -1. \quad (12.12)$$

These statements concern entire infinite families of inequalities and asymptotic regimes, rather than additional finite entries in an OEIS table. The two ends of the same measures are quite different: an irrational power law at zero and a logarithmically Poisson-scale tail at infinity.

12.3 Attribution and limitations

The sequence definitions and reversion identities are established OEIS material [1, 5]. Bound (12.3) is the theorem of Part I, not a new theorem of Part II. Fock-space models for noncrossing moments are standard [6]; so are the general implications from Stieltjes moments to infinite log-convexity [7] and the relevant Tauberian principle [8]. We give the particular proofs needed, including positivity of the operators and the Tauberian constant.

The contributions developed here are the application of these constructions to this nonlinear fixed point, its determinate positive-measure realization, the exact negative-axis equations and certified approximants, the constant-one golden-ratio laws, and the ratio and tail deductions. The selected repository report (Part I) and targeted searches did not reveal these specific conclusions. That is a bounded novelty audit, not an exhaustive priority determination. The new statements are unrefereed written proofs; no formal verification is claimed. We do not prove absolute continuity of either measure, and do not settle the earlier finer Bell-number conjecture.

13 Two operator constructions that preserve positivity

This section supplies the central existence mechanism. All operators here are bounded: input measures have compact support. Unbounded support is introduced only by taking weak limits in Section 14.

13.1 The renewal transformation

Lemma 13.1 (A positive block operator). *Let ξ be a compactly supported probability measure on $[0, \infty)$, and write $B(z) = \int (1 - zt)^{-1} d\xi(t)$ near zero. Then*

$$R(z) = \frac{1}{1 - zB(z)} \quad (13.1)$$

is the moment series of a compactly supported probability measure on $[0, \infty)$.

Proof. On $K = \mathbb{C} \oplus L^2(\xi)$ let T be multiplication by t and put

$$Q = \begin{pmatrix} 1 & \langle \sqrt{t}, \cdot \rangle \\ \sqrt{t} & T \end{pmatrix}.$$

For $(u, v) \in K$, its quadratic form is

$$\langle (u, v), Q(u, v) \rangle = \int |u + \sqrt{t} v(t)|^2 d\xi(t) \geq 0.$$

The mass-one assumption on ξ is used in the coefficient of $|u|^2$. Thus Q is bounded, self-adjoint and positive. Its spectral measure in the unit vector $e_0 = (1, 0)$ is a compactly supported probability measure on $[0, \infty)$. A Schur complement, initially for sufficiently small z , gives its moment series as

$$\begin{aligned} \langle e_0, (I - zQ)^{-1} e_0 \rangle &= \frac{1}{1 - z - z^2 \int t(1 - zt)^{-1} d\xi(t)} \\ &= \frac{1}{1 - zB(z)}. \end{aligned}$$

The identity $B(z) = 1 + z \int t(1 - zt)^{-1} d\xi(t)$ proves the last step. □

In coefficients, if $B = \sum b_n z^n$ and $R = \sum d_n z^n$, then

$$d_0 = 1, \quad d_n = \sum_{j=0}^{n-1} b_j d_{n-1-j}. \quad (13.2)$$

Positivity of these coefficients alone would not prove the moment property; the positive operator is the stronger ingredient.

13.2 From a positive moment sequence to noncrossing moments

Let $\text{NC}(n)$ denote the partitions of $\{1, \dots, n\}$ with no two distinct blocks containing $i < j < k < \ell$ with i, k in one block and j, ℓ in the other. These are the noncrossing partitions. The empty partition has weight 1.

Lemma 13.2 (A positive full-Fock operator). *Suppose $d_n = \int t^n d\rho(t)$, where ρ is a compactly supported probability measure on $[0, \infty)$. Put*

$$m_0 = 1, \quad m_n = \sum_{\pi \in \text{NC}(n)} \prod_{V \in \pi} d_{|V|}. \quad (13.3)$$

Then (m_n) is also a compactly supported Stieltjes moment sequence. Writing $R(z) = \sum d_n z^n$, its moment series is the unique formal solution

$$M(z) = R(zM(z)), \quad M(0) = 1. \quad (13.4)$$

Proof. Set $H = L^2(\rho)$ and let T be multiplication by t . The full Fock space is

$$\mathcal{F}(H) = \mathbb{C}\Omega \oplus \bigoplus_{k \geq 1} H^{\otimes k}.$$

The vector Ω is the vacuum. Define maps to $H \otimes \mathcal{F}(H)$ by

$$D\Omega = 0, \quad D(h_1 \otimes \cdots \otimes h_k) = h_1 \otimes (h_2 \otimes \cdots \otimes h_k), \quad J\psi = 1 \otimes \psi.$$

The empty tail means Ω . Both maps have norm 1, and

$$X = (D + J)^*(T \otimes I)(D + J) \geq 0. \quad (13.5)$$

It is bounded. Expanding gives four operations: multiplication by T on the first tensor factor of a nonempty word; creation of a first factor t ; annihilation by the linear functional $\langle t, \cdot \rangle$; and $d_1 I$. The gauge operation vanishes on Ω .

We verify the vacuum moments, not merely positivity. Expand X^n into words in these four operations and apply them from right to left. A term contributing to the vacuum never attempts to remove a factor from an empty word and ends at tensor length zero. Match each created tensor factor with its eventual annihilation, using the last-in, first-out order of first tensor factors. Gauges acting on that factor, together with its creation and annihilation, form a block. Scalar operations form singleton blocks. Operations acting on factors above it form nested blocks. This correspondence is a bijection with noncrossing partitions; reading such a partition from one end reconstructs the operator choices and the stack of active factors.

A block of size $r \geq 2$ has weight $\langle t, T^{r-2}t \rangle = \int t^r d\rho(t) = d_r$; a singleton has weight d_1 . Nested closed pieces leave the lower tensor factors untouched and contribute their own scalar weights. Thus $\langle \Omega, X^n \Omega \rangle$ is exactly (13.3). The positive spectral measure of X at Ω proves the assertion.

Finally, consider the block containing the first point of a noncrossing partition. If its size is r , its weight is d_r , and there are r independent intervening or final intervals, each carrying a noncrossing partition. Consequently

$$M(z) = 1 + \sum_{r \geq 1} d_r z^r M(z)^r = R(zM(z)).$$

The degree- n equation determines m_n from earlier m_j , proving uniqueness. $\square$

For example, $m_1 = d_1$, $m_2 = d_1^2 + d_2$, and $m_3 = d_1^3 + 3d_1 d_2 + d_3$. In free-probability terminology, this is the rate-one free compound Poisson construction with jump law ρ ; the operator proof above is sufficient without adopting that terminology.

13.3 The nonlinear transform

Combining the two lemmas gives the following general result.

Theorem 13.3 (Moment-preserving nonlinear transform). *For a compact Stieltjes moment series B with $B(0) = 1$, the unique formal solution $M = \mathcal{T}(B)$ of*

$$M(z) = 1 + zM(z)^2B(zM(z)), \quad M(0) = 1, \quad (13.6)$$

is again a compact Stieltjes moment series.

Proof. Apply Lemma 13.1 to obtain $R = (1 - zB)^{-1}$, and then Lemma 13.2 to obtain $M = R(zM)$. The resulting identity is $M = (1 - zMB(zM))^{-1}$, which is equivalent to (13.6). $\square$

Corollary 13.4 (Removal of compact support). *The same transformation preserves normalized Stieltjes moment sequences with all moments finite, without a compact-support assumption.*

Proof. Replace an input random variable $X \geq 0$ by $\min(X, R)$. Its moments converge to the input moments. Every output moment in the two lemmas is a fixed polynomial in finitely many input moments, so the output moments converge as well. The output first moment is 1, and bounds on each next moment give tightness and uniform integrability. A weak subsequential limit therefore represents all output moments. No assertion of determinacy is needed for this preservation statement. $\square$

This is a preservation statement on an entire class of series. The special feature of A088714 is that it is the coefficientwise attracting fixed point of this transformation.

14 The fixed point as a determinate moment law

14.1 Exact coefficient stabilization

Start with $B_0(z) = 1$, the moment series of δ_0 , and define

$$B_{r+1} = \mathcal{T}(B_r), \quad C_r = (1 - zB_r)^{-1}. \quad (14.1)$$

Let μ_r, ν_r be the compact probability measures supplied above for B_r, C_r , respectively. All have nonnegative support.

Lemma 14.1 (One additional correct coefficient per step). *If $B - \tilde{B}$ is divisible by z^N , $N \geq 1$, then $\mathcal{T}(B) - \mathcal{T}(\tilde{B})$ is divisible by z^{N+1} . In particular,*

$$B_r \equiv A \pmod{z^{r+1}}, \quad C_r \equiv C \pmod{z^{r+2}}. \quad (14.2)$$

Proof. For $M = \mathcal{T}(B)$, the coefficient m_n , $n \geq 1$, depends only on $b_0, \dots, b_{n-1}$ and $m_0, \dots, m_{n-1}$. Induction on $n \leq N$ therefore proves agreement. Since B_0 and A agree at degree zero, iteration proves the first congruence. Equation (13.2) proves the second. $\square$

14.2 Existence and uniqueness of the limiting measures

The first moment of μ_r is 1 for $r \geq 1$. Hence $\mu_r([R, \infty)) \leq 1/R$; the measures are tight. Every subsequential weak limit has all moments a_n . Indeed, for fixed n , the moments of orders n and $n+1$ eventually stabilize, and

$$\int_{[R, \infty)} t^n d\mu_r(t) \leq \frac{a_{n+1}}{R}$$

for sufficiently large r . Bounded continuous cutoffs, followed by $R \rightarrow \infty$, justify passage of the n th moment to the limit. Denote one such limit by μ .

Bound (12.3) and $\log(n!) = n \log n - n + O(\log n)$ imply

$$\left(\frac{a_n}{n!}\right)^{1/n} \rightarrow 0. \quad (14.3)$$

Thus for every $v > 0$, monotone convergence gives

$$\int e^{vt} d\mu(t) = \sum_{n \geq 0} \frac{a_n v^n}{n!} < \infty.$$

The same argument applies to any measure with these moments. Its characteristic function is then the absolutely justified moment expansion $\sum a_n (iu)^n / n!$. Equality of characteristic functions determines the probability measure. Thus μ is unique, and the whole sequence μ_r converges weakly to it.

For ν_r , the first moment is again 1, and its moments converge coefficientwise to c_n . Tightness and uniform integrability give a representing measure ν . To prove uniqueness, observe first that the moments of a nonnegative random variable satisfy

$$a_i a_j \leq a_{i+j}. \quad (14.4)$$

For example, take independent copies X, Y with law μ and use $\mathbb{E}[(X^i - Y^i)(X^j - Y^j)] \geq 0$. Also $a_j \leq a_{j+1}$: log-convexity of moments follows from Cauchy–Schwarz, and the initial ratio is $a_1/a_0 = 1$. Expanding the geometric series in (12.2), for $n \geq 1$ we obtain

$$c_n = \sum_{k=1}^n \sum_{\substack{s_1 + \dots + s_k = n \\ s_i \geq 1}} \prod_{i=1}^k a_{s_i-1}. \quad (14.5)$$

Each product is at most $a_{n-k} \leq a_{n-1}$, and there are 2^{n-1} compositions of n . Therefore

$$a_{n-1} \leq c_n \leq 2^{n-1} a_{n-1}. \quad (14.6)$$

It follows that $\sum c_n v^n / n!$ is entire, proving determinacy and all positive exponential moments for ν as well. Consequently $\nu_r \Rightarrow \nu$.

Finally, (12.3) gives $a_n^{1/n} \rightarrow \infty$. The moments of a probability measure supported in $[0, R]$ are at most R^n . Thus μ has unbounded support. Inequality (14.6) proves the same for ν .

15 Strict Hankel positivity and iterated log-convexity

We provide the strict statements explicitly, since existence of a representing measure alone only guarantees nonnegative minors in general.

Lemma 15.1 (Generalized Vandermonde positivity). *If $0 \leq e_1 < \cdots < e_k$ are integers and $0 < t_1 < \cdots < t_k$, then $\det[t_j^{e_i}]_{i,j=1}^k > 0$.*

Proof. A nonzero linear combination of k distinct nonnegative powers has at most $k - 1$ distinct positive zeros. This follows by induction: divide by the smallest power, differentiate, and use Rolle's theorem. Consequently the determinant cannot vanish in the connected chamber $0 < t_1 < \cdots < t_k$, so its sign is constant. Letting the nodes approach a common positive point identifies its sign with that of the Wronskian

$$t^{\sum e_i - k(k-1)/2} \prod_{i < j} (e_j - e_i) > 0$$

times a positive ordinary Vandermonde factor. This proves the sign. $\square$

Theorem 15.2 (Strict total positivity). *Let m_n be the moments of a probability measure σ on $[0, \infty)$ with all moments finite and infinitely many positive support points. For any strictly increasing nonnegative index lists $i_1 < \cdots < i_k$ and $j_1 < \cdots < j_k$,*

$$\det[m_{i_p+j_q}]_{p,q=1}^k > 0. \quad (15.1)$$

In particular this holds for both (a_n) and (c_n) .

Proof. Expanding two determinants and integrating termwise gives

$$\det[m_{i_p+j_q}] = \frac{1}{k!} \int \cdots \int \det[t_\ell^{i_p}]_{p,\ell} \det[t_\ell^{j_q}]_{q,\ell} \prod_{\ell=1}^k d\sigma(t_\ell). \quad (15.2)$$

All terms are integrable since all moments are finite. The product is unchanged by a permutation of the t_ℓ , and is nonnegative everywhere: on distinct positive points this follows from Lemma 15.1; on repeated points it vanishes, and at zero it follows by continuity. Choose k separated positive support points and disjoint small ordered neighborhoods. Each neighborhood has positive mass, and the integrand is strictly positive on their product. Thus the integral is strictly positive. Both measures constructed above have unbounded support, so the hypothesis holds for every k . $\square$

Theorem 15.3 (Strict infinite log-convexity). *For $m = a$ or $m = c$, every $\mathcal{L}^k m$, $k \geq 1$, is a Stieltjes moment sequence with an unbounded positive support representation. In particular, $(\mathcal{L}^k m)_n > 0$ for every $n \geq 0$.*

Proof. If $m_n = \int t^n d\sigma(t)$, then

$$m_n m_{n+2} - m_{n+1}^2 = \frac{1}{2} \int \int (xy)^n (x - y)^2 d\sigma(x) d\sigma(y). \quad (15.3)$$

Thus $\mathcal{L}m$ is the moment sequence of the pushforward under $(x, y) \mapsto xy$ of the finite positive measure $\frac{1}{2}(x - y)^2 d\sigma(x) d\sigma(y)$. All moments remain finite. If σ has unbounded support and a positive support point y_0 , choose support points $x_j \rightarrow \infty$. Small product neighborhoods of (x_j, y_0) have positive weighted mass and yield arbitrarily large positive product support points. Therefore the new measure also has unbounded support and positive total mass. Repeat this argument. The representing measures after the first iteration need not have mass one, but identity (15.3) does not require normalization. $\square$

The general preservation phenomenon is known [7]; its use here is justified by the newly constructed representing measures. Ordinary strict log-convexity, $a_n a_{n+2} > a_{n+1}^2$, is only the first layer.

15.1 An exact determinant bridge between the two sequences

For a sequence m , write $D_n^{(h)}(m) = \det[m_{i+j+h}]_{i,j=0}^{n-1}$, with $D_0^{(h)} = 1$. The following identities explain the repeated determinants in the numerical tables.

Proposition 15.4 (Renewal Hankel identities). *For every $n \geq 1$,*

$$D_n^{(1)}(c) = D_n^{(0)}(a), \quad D_n^{(0)}(c) = D_{n-1}^{(1)}(a). \quad (15.4)$$

These are algebraic consequences of $C = (1 - zA)^{-1}$, independent of moment positivity.

Proof. For a series $M = \sum m_n z^n$, its two bivariate Hankel kernels are

$$H_M(x, y) = \frac{xM(x) - yM(y)}{x - y}, \quad S_M(x, y) = \frac{M(x) - M(y)}{x - y}.$$

They have coefficients m_{i+j} and m_{i+j+1} , respectively. A direct substitution gives

$$S_C(x, y) = C(x)C(y)H_A(x, y), \quad H_C(x, y) = C(x)C(y)(1 + xyS_A(x, y)).$$

Multiplication of a bivariate kernel by $C(x)C(y)$ is congruence by a lower-triangular Toeplitz matrix with diagonal entries $c_0 = 1$. Its finite leading determinants are unchanged. The first kernel identity proves the first formula; in the second identity the inside leading matrix is the block diagonal matrix with blocks 1 and the leading $(n-1) \times (n-1)$ shifted Hankel matrix of a . This proves the second. $\square$

16 Exact functions behind the divergent series

16.1 Passing the nonlinear identity to the limit

Let

$$F_r(s) = \int \frac{d\mu_r(t)}{1 + st}, \quad G_r(s) = \int \frac{d\nu_r(t)}{1 + st}.$$

For the compact measures, the formal identities become analytic identities near $s = 0$:

$$F_{r+1}(s) = 1 - sF_{r+1}(s)^2 F_r(sF_{r+1}(s)), \quad G_r(s) = \frac{1}{1 + sF_r(s)}. \quad (16.1)$$

They hold for every $s > 0$. Indeed, both sides are real-analytic on this connected interval, their composed arguments remain positive, and they agree on an initial interval. This continuation argument concerns compact approximants, not the divergent limiting power series.

Weak convergence gives $F_r(s) \rightarrow F(s)$ and $G_r(s) \rightarrow G(s)$ for each $s > 0$. It also permits the variable-argument passage in (16.1). For example, on any interval $s \geq \delta > 0$,

$$|F'_r(s)| = \int \frac{t}{(1 + st)^2} d\mu_r(t) \leq \frac{1}{4\delta}.$$

Since $sF_{r+1}(s) \rightarrow sF(s) > 0$, this equicontinuity bound and pointwise convergence imply $F_r(sF_{r+1}(s)) \rightarrow F(sF(s))$. Thus (12.7) follows.

16.2 The inverse equation and absence of an atom

For every $s > 0$, $0 < F(s) < 1$ because μ has positive mass away from zero. Moreover

$$U'(s) = \int \frac{d\mu(t)}{(1+st)^2} > 0, \quad U(s) < s. \quad (16.2)$$

The first identity in (12.7) is equivalent to

$$s = U(s)(1 + U(U(s))). \quad (16.3)$$

If U were bounded, its right side would be bounded, which is impossible. As $U(0+) = 0$, strict monotonicity proves it is a homeomorphism onto $(0, \infty)$, and (12.8) follows.

Also $F(s) \downarrow \mu(\{0\})$. A positive limit p would give $sF(s)^2F(sF(s)) \geq sp^3$, contradicting (12.7). Hence $F(s) \rightarrow 0$. Since $U(s) \rightarrow \infty$, $G(s) = (1 + U(s))^{-1} \rightarrow 0$ as well. This proves the absence of atoms at zero for both measures.

16.3 Asymptotic series with explicit remainders

The identity for a finite geometric sum gives, for every integer $N \geq 0$ and every $s > 0$,

$$F(s) = \sum_{j=0}^N (-1)^j a_j s^j + (-1)^{N+1} s^{N+1} \int \frac{t^{N+1}}{1+st} d\mu(t). \quad (16.4)$$

In particular, the remainder has its displayed sign and absolute value at most $a_{N+1}s^{N+1}$. The analogous statement holds for G with c_j . Thus the alternating formal series is a genuine Poincaré asymptotic expansion as $s \downarrow 0$, with finite-order upper and lower bounds, even though its infinite sum has radius zero.

The entire functions

$$E_a(z) = \sum_{n \geq 0} \frac{a_n z^n}{n!}, \quad E_c(z) = \sum_{n \geq 0} \frac{c_n z^n}{n!}$$

are the exponential moment transforms of μ, ν . Consequently

$$F(s) = \int_0^\infty e^{-v} E_a(-sv) dv, \quad G(s) = \int_0^\infty e^{-v} E_c(-sv) dv \quad (s > 0). \quad (16.5)$$

To justify this, first use exponential integrability to identify $E_a(-u) = \int e^{-ut} d\mu(t)$, and then use Tonelli's theorem in (16.5). The integrand $E_a(-sv)$ lies between 0 and 1. This is a Borel–Laplace representation on the negative real axis. It means integrating the *whole entire function*; exchanging its infinite series with this integral is not justified and would recover the divergent ordinary series.

These constructions select canonical transforms from the moment problem. They do not assert uniqueness among every possible smooth, non-Stieltjes solution of the nonlinear equation.

17 Certified algebraic approximants and Fibonacci exponents

17.1 Alternating pointwise bounds

Proposition 17.1 (Order reversal). *Given the Stieltjes transform H of a probability measure on $[0, \infty)$, the equation*

$$y + sy^2 H(sy) = 1 \quad (17.1)$$

has exactly one solution in $(0, 1)$ for each $s > 0$. Increasing H pointwise decreases this solution pointwise.

Proof. The left side is continuous, is 0 at $y = 0$, exceeds 1 at $y = 1$, and is strictly increasing. For the last assertion, each function $y^2/(1 + sty)$ has derivative $y(2 + sty)/(1 + sty)^2 > 0$ on $y > 0$, so its average is increasing. If $H \geq \tilde{H}$, the left side for H is at least that for $\tilde{H}$ at every y , and its unique crossing occurs no later. $\square$

Since $F_0 = 1 \geq F$, repeated order reversal in (16.1) gives

$$F_1 \leq F_3 \leq F_5 \leq \cdots \leq F \leq \cdots \leq F_4 \leq F_2 \leq F_0 \quad \text{on } (0, \infty). \quad (17.2)$$

For nesting, start with $F_2 \leq F_0$ and apply the same reversal successively. Section 14 proves that both bounding sequences converge to F . The convergence is uniform on compact subintervals of $(0, \infty)$ by monotonicity and continuity.

Every B_r is algebraic. Indeed, if C_r satisfies a nonzero polynomial relation $P(z, C_r(z)) = 0$, then $B_{r+1} = C_r(zB_{r+1})$ satisfies $P(zB_{r+1}, B_{r+1}) = 0$. The substitution $z^i y^j \mapsto z^i y^{i+j}$ is injective on polynomials, so this relation is nonzero. Start with $B_0 = 1$ and use $C_r = (1 - zB_r)^{-1}$.

The first two bounds are especially simple:

$$\frac{2}{1 + \sqrt{1 + 4s}} = F_1(s) \leq F(s) \leq F_2(s), \quad F_2(s) + sF_2(s)^3 = 1. \quad (17.3)$$

The upper bound is the unique positive root of the cubic. To derive it, write $v = F_1(sy)$, so that $v + syv^2 = 1$ and $1 - y = sy^2v$; eliminating v gives $1 - y = sy^3$. Therefore (17.2) is also an effective route to certified bounds using algebraic root isolation or nested interval bisection, without truncating a divergent ordinary series at large s .

17.2 The Fibonacci pattern

Let $f_0 = 0$, $f_1 = 1$, and $f_{r+2} = f_{r+1} + f_r$ be the Fibonacci numbers.

Theorem 17.2 (Endpoint exponents of every approximant). *For each fixed $r \geq 0$,*

$$F_r(s) \sim s^{-\alpha_r}, \quad \alpha_r = \frac{f_r}{f_{r+2}}, \quad s \rightarrow \infty. \quad (17.4)$$

Every leading constant is 1. The exponents alternate around α and converge to it:

$$0, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{2}{5}, \quad \frac{3}{8}, \quad \frac{5}{13}, \quad \frac{8}{21}, \dots \longrightarrow \alpha.$$

Proof. Put $U_r(s) = sF_r(s)$. The finite identity yields

$$U_{r+1}^{-1}(t) = t(1 + U_r(t)), \quad U_0(t) = t.$$

This inverse relation also proves inductively that every U_r is an increasing homeomorphism onto $(0, \infty)$. If $U_r(t) \sim t^{b_r}$, then its successor's inverse is asymptotic to t^{1+b_r} , so $U_{r+1}(s) \sim s^{1/(1+b_r)}$. With $b_0 = 1$, this gives $b_r = f_{r+1}/f_{r+2}$, including the constant 1. Since $\alpha_r = 1 - b_r$, the formula follows. The elementary recurrence $b \mapsto 1/(1 + b)$ has fixed point β and reverses the order around it, giving the stated alternating convergence. $\square$

The brackets and this theorem alone determine $\log F(s)/\log s \rightarrow -\alpha$. They do *not* justify the full constant-one equivalent by interchanging $r \rightarrow \infty$ with $s \rightarrow \infty$. The next section supplies a separate rigidity argument that closes precisely this gap.

18 Why the golden-ratio law has leading constant one

The derivation below avoids assuming regular variation or differentiable asymptotics. Its conclusion is stronger than simply guessing a power in the nonlinear equation.

Lemma 18.1 (An asymptotic iteration rigidity principle). *Suppose $g : \mathbb{R} \rightarrow \mathbb{R}$ is increasing and continuous, $g(x) < x$, $g(x) \rightarrow \infty$ as $x \rightarrow \infty$, and*

$$g(g(x)) + g(x) = x - \varepsilon(x), \quad \varepsilon(x) \rightarrow 0. \quad (18.1)$$

Then $g(x) - \beta x \rightarrow 0$, where $\beta^2 + \beta = 1$ and $\beta > 0$.

Proof. Put $d(x) = g(x) - \beta x$. Substitution in (18.1) gives

$$d(x) = -\beta d(g(x)) - \beta \varepsilon(x). \quad (18.2)$$

Fix a sufficiently large real X . For $x > X$, iterate g until it first reaches a point at most X , and let $N = N(x)$ be that number of steps. It is finite: otherwise the decreasing orbit has a limit at least X , and continuity makes that limit a fixed point, contrary to $g(y) < y$. By monotonicity, the exit point lies in $[g(X), X]$. Also $N(x) \rightarrow \infty$ as $x \rightarrow \infty$, because every fixed finite iterate of g tends to infinity.

Let $M_X = \sup_{[g(X), X]} |d|$ and $E_X = \sup_{y \geq X} |\varepsilon(y)|$. Iterating (18.2) gives

$$|d(x)| \leq \beta^{N(x)} M_X + \frac{\beta}{1 - \beta} E_X.$$

Take the limit superior as $x \rightarrow \infty$, then let $X \rightarrow \infty$. The result is zero, proving the lemma. $\square$

Apply this lemma to

$$g(x) = \log U(e^x). \quad (18.3)$$

The properties of U prove all its monotonicity, continuity and range hypotheses. From (16.3),

$$U(U(s)) = \frac{s}{U(s)} - 1 = \frac{s}{U(s)} (1 - F(s)).$$

Taking logarithms at $s = e^x$ yields

$$g(g(x)) + g(x) = x + \log(1 - F(e^x)).$$

Thus $\varepsilon(x) = -\log(1 - F(e^x)) \rightarrow 0$, since $F(e^x) \rightarrow 0$. Lemma 18.1 now gives

$$\log \frac{U(s)}{s^\beta} \rightarrow 0.$$

Exponentiating proves $U(s) \sim s^\beta$, hence $F(s) = U(s)/s \sim s^{-\alpha}$. Finally $G(s) = (1 + U(s))^{-1} \sim s^{-\beta}$. This proves (12.9), with no unspecified multiplicative constant and no exchange of the two limits in the preceding section.

19 Recovering the measures near zero

For completeness we prove the precise special case of the Stieltjes Tauberian theorem used here. General versions and their history are available in [8]. The normalization of the transform matters.

Lemma 19.1 (Power-law Stieltjes Tauberian lemma). *Let σ be a probability measure on $[0, \infty)$ and $H(s) = \int (1 + st)^{-1} d\sigma(t)$. If*

$$H(s) \sim K s^{-\gamma} \quad (s \rightarrow \infty), \quad K > 0, \quad 0 < \gamma < 1,$$

then

$$\sigma([0, t]) \sim K \frac{\sin(\pi\gamma)}{\pi\gamma} t^\gamma \quad (t \downarrow 0). \quad (19.1)$$

Proof. Let η_s be the probability measure of the rescaled variable $y = st$, weighted by $(1 + st)^{-1}/H(s)$. Regard it as a measure on the compact interval $[0, 1]$ using the coordinate $w = y/(1 + y)$, with $y = \infty$ corresponding to $w = 1$. For $u > 0$,

$$\int \frac{1 + y}{1 + uy} d\eta_s(y) = \frac{H(us)}{H(s)} \rightarrow u^{-\gamma}. \quad (19.2)$$

The integrand is the bounded continuous function $[1 + (u - 1)w]^{-1}$ on $[0, 1]$.

Consider the probability measure on $(0, \infty)$ with density

$$\frac{\sin(\pi\gamma)}{\pi} \frac{y^{\gamma-1}}{1 + y}. \quad (19.3)$$

Euler's beta integral $\int_0^\infty y^{\gamma-1}(1 + y)^{-1} dy = \pi/\sin(\pi\gamma)$ normalizes it. Substitution $v = uy$ shows that its integral in (19.2) is $u^{-\gamma}$. This probability measure is the unique one on the compact w -interval with those integrals. Indeed, for $|u - 1| < 1$ expand $[1 + (u - 1)w]^{-1}$ uniformly as a power series. The integrals determine all moments of w ; polynomials are dense in continuous functions on $[0, 1]$, so these moments determine the measure.

Every subsequential weak limit of η_s therefore equals (19.3). Compactness proves convergence of the whole family. In particular, the limit has no mass at $y = 1$ or at either compactification endpoint. The bounded function $(1 + y)\mathbf{1}_{\{y \leq 1\}}$ is continuous except at $y = 1$, so weak convergence gives

$$\frac{\sigma([0, 1/s])}{H(s)} = \int_{y \leq 1} (1 + y) d\eta_s(y) \rightarrow \frac{\sin(\pi\gamma)}{\pi} \int_0^1 y^{\gamma-1} dy = \frac{\sin(\pi\gamma)}{\pi\gamma}.$$

Using the assumed equivalent for $H(s)$ proves (19.1). $\square$

Apply the lemma to F with $(K, \gamma) = (1, \alpha)$ and to G with $(K, \gamma) = (1, \beta)$. This proves (12.10). In particular, arbitrarily small positive intervals adjacent to zero have positive mass. The result is about *distribution functions*: it does not justify differentiating the equivalent or asserting a density.

Corollary 19.2 (Sharp negative-moment thresholds). *For $q > 0$,*

$$\int t^{-q} d\mu(t) < \infty \iff q < \alpha, \quad \int t^{-q} d\nu(t) < \infty \iff q < \beta.$$

At the critical exponent γ for either law σ ,

$$\int_{[\varepsilon, 1]} t^{-\gamma} d\sigma(t) \sim \frac{\sin(\pi\gamma)}{\pi} \log \frac{1}{\varepsilon}. \quad (19.4)$$

Proof. For $0 < x \leq 1$, $x^{-q} = 1 + q \int_x^1 t^{-q-1} dt$. Tonelli's theorem reduces finiteness near zero to that of $\int_0^1 t^{-q-1} \sigma([0, t]) dt$. The distribution-function equivalent makes this the elementary integral of $t^{\gamma-q-1}$. The part on $[1, \infty)$ is bounded by the total mass. At $q = \gamma$, integration by parts on $[\varepsilon, 1]$ gives a bounded endpoint term plus $\gamma \int_\varepsilon^1 t^{-\gamma-1} \sigma([0, t]) dt$. Its equivalent is the asserted logarithm. Possible endpoint atoms do not alter it: the distribution-function equivalent makes their scaled contribution bounded, hence negligible relative to the logarithm. $\square$

20 From global growth to consecutive-coefficient asymptotics

The earlier root equivalent by itself cannot control neighboring ratios. The moment construction supplies the missing convexity.

Lemma 20.1 (A convex-slope principle). *Suppose $m_n > 0$ is log-convex and*

$$\log m_n = n \log n - n \log \log n - n + o(n). \quad (20.1)$$

Then $m_{n+1}/m_n \sim n/\log n$.

Proof. Let $h(n) = \log m_n$, and interpolate linearly between integers. Log-convexity makes this interpolation convex. Fix $0 < \delta < 1$. The forward slope $h(n+1) - h(n)$ is bounded below by the secant from $\lfloor(1-\delta)n\rfloor$ to n , and above by the secant from n to $\lceil(1+\delta)n\rceil$. Inserting (20.1), accounting for rounding by $o(1)$ in the quotients, gives

$$\begin{aligned} \liminf_{n \rightarrow \infty} (h(n+1) - h(n) - \log n + \log \log n) &\geq -\frac{1-\delta}{\delta} \log(1-\delta) - 1, \\ \limsup_{n \rightarrow \infty} (h(n+1) - h(n) - \log n + \log \log n) &\leq \frac{1+\delta}{\delta} \log(1+\delta) - 1. \end{aligned}$$

Both right sides tend to zero as $\delta \downarrow 0$. Hence the expression inside parentheses tends to zero. Exponentiation proves the result. $\square$

Strict log-convexity from Section 15 and (12.3) immediately prove the first equivalent in (12.11). It remains to transfer the sharp logarithmic growth to c_n ; the factor 2^{n-1} in (14.6) is too large for this purpose.

Proposition 20.2 (Sharper comparison for the companion sequence). *As $n \rightarrow \infty$,*

$$\log c_n = \log a_{n-1} + O(\log n) = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (20.2)$$

Consequently $c_{n+1}/c_n \sim n/\log n$.

Proof. Keeping track of the number of parts in (14.5), and using (14.4), gives

$$a_{n-1} \leq c_n \leq \sum_{j=0}^{n-1} \binom{n-1}{j} a_j. \quad (20.3)$$

Write $r_j = a_{j+1}/a_j$, which is increasing and satisfies $r_j \sim j/\log j$. Put $m = \lfloor(n-1)/2\rfloor$. The part $j \leq m$ of the upper sum is at most $2^{n-1} a_m = o(a_{n-1})$ by (12.3). For $j > m$, monotonicity of r_j gives

$$a_j \leq a_{n-1} r_m^{-(n-1-j)}.$$

Thus the rest of the sum is bounded by

$$a_{n-1}(1 + 1/r_m)^{n-1} \leq a_{n-1} \exp((n-1)/r_m) = a_{n-1} \exp(O(\log n)).$$

Together with the lower bound this proves the first equality in (20.2). The second follows from (12.3) and $\log r_{n-1} = O(\log n)$. Finally apply Lemma 20.1 to c , whose log-convexity has already been proved. $\square$

Corollary 20.3 (A quantitative, though slow, ratio estimate). *For $m = a$ or $m = c$,*

$$\frac{m_{n+1}}{m_n} = \frac{n}{\log n} \left(1 + O \left(\sqrt{\frac{\log \log n}{\log n}} \right) \right).$$

Proof. Let $h_0(t) = t \log t - t \log \log t - t$ and $e_n = \log \log n / \log n$. On $[n/2, 2n]$, the available logarithmic estimate has error $O(ne_n)$, uniformly. For secants over length δn , this contributes $O(e_n/\delta)$. Since $h_0''(t) = O(1/n)$ there, the secant differs from $h_0'(n)$ by $O(\delta)$; also $h_0'(n) = \log n - \log \log n + O(1/\log n)$. Take $\delta = \sqrt{e_n}$ in the convex-slope bounds. Rounding endpoints contributes only $O(\log n/(\delta n))$. Exponentiation gives the result. $\square$

This is not a proof that $c_n \sim a_{n-1}$. The polynomial-factor comparison is sufficient for the asserted logarithmic and ratio laws, and no stronger equivalent is used below.

21 The far tails from size-biased concentration

We prove a general lemma and then apply it to both measures. This keeps the lower-tail-bound argument separate from an insufficient Markov upper bound.

Lemma 21.1 (A moment-growth tail principle). *Let σ be a probability measure on $[0, \infty)$ with positive moments m_n , unbounded support, and*

$$\log m_n = n \log n - n \log \log n - n + o(n), \quad r_n := m_{n+1}/m_n \sim n/\log n.$$

Then $\log \sigma([x, \infty)) \sim -x \log x$.

Proof. For the upper bound, Markov's inequality gives $\sigma([x, \infty)) \leq m_n/x^n$. Choose $n = \lfloor x \log x \rfloor$. Then $\log n - \log \log n = \log x + o(1)$, and consequently

$$\log \sigma([x, \infty)) \leq -(1 + o(1))x \log x. \quad (21.1)$$

For the reverse inequality define the probability measure

$$d\sigma_n(t) = \frac{t^n}{m_n} d\sigma(t).$$

Its mean is r_n and its relative variance is

$$\frac{\text{Var}_{\sigma_n}(t)}{r_n^2} = \frac{r_{n+1}}{r_n} - 1 \longrightarrow 0.$$

The last limit follows directly from the ratio equivalent at n and $n+1$. Thus for each fixed $0 < \varepsilon < 1$,

$$\sigma_n([(1-\varepsilon)r_n, (1+\varepsilon)r_n]) \longrightarrow 1. \quad (21.2)$$

Fix $\delta > 0$ and choose $\varepsilon > 0$ so small that $(1 - \varepsilon)(1 + \delta) > 1$. Now take $n = \lfloor (1 + \delta)x \log x \rfloor$. Then $r_n \sim (1 + \delta)x$, so the interval in (21.2) lies in $[x, \infty)$ for large x . On it, $t^n \leq ((1 + \varepsilon)r_n)^n$, which implies

$$\sigma([x, \infty)) \geq \frac{m_n}{((1 + \varepsilon)r_n)^n} \sigma_n([(1 - \varepsilon)r_n, (1 + \varepsilon)r_n]).$$

Taking logarithms and using both hypotheses gives

$$\liminf_{x \rightarrow \infty} \frac{\log \sigma([x, \infty))}{x \log x} \geq -(1 + \delta)(1 + \log(1 + \varepsilon)).$$

Let $\delta \downarrow 0$ with admissible $\varepsilon \downarrow 0$, and combine with (21.1). This proves the limit. $\square$

Apply the lemma to μ using (12.3) and to ν using (20.2), with their already proved ratio laws. This proves Theorem 12.3. Equivalently, for every fixed $\varepsilon > 0$, for sufficiently large x each tail lies between $\exp(-(1 + \varepsilon)x \log x)$ and $\exp(-(1 - \varepsilon)x \log x)$. Nothing here supplies a multiplicative tail equivalent or a density.

22 Computational audits and certified orbit enclosures

22.1 Exact calculations actually performed

The computations are checks of implementations and finite instances, not substitutes for the all-index proofs. The file `code/02-golden-moments-verify.py` uses only the Python standard library and performs the following tasks. It independently generates $a_0, \dots, a_{401}$ by the triangular recurrence (2.8) of Part I (whose $c_{n,k}$ are not the c_n of A088713), obtains $c_0, \dots, c_{401}$ by (13.2), and checks all 21 displayed A088714 terms and all 22 displayed A088713 terms retrieved from the OEIS entries.

Two different formal calculations check the defining functional equation and the noncrossing relation $A = C(zA)$ through degree 30. The compact transform iteration is separately checked through stage 16, including its exact stabilization property. A direct rational-arithmetic implementation of the full-Fock operator for $\rho = \frac{1}{2}(\delta_1 + \delta_3)$ agrees with the noncrossing formula through moment 10. In that calculation the tensor factors are functions on $\{1, 3\}$, with their weighted inner product; no square-root or floating-point approximation enters.

For both sequences, every minor of order 1 through 4 whose row and column indices lie in $\{0, \dots, 7\}$ was checked: 17,768 minors in total, all strictly positive. An exact fraction-free determinant algorithm also checks all 40 leading and once-shifted determinants of orders 1 through 10 and all 20 determinant identities in Proposition 15.4 for those orders. Six iterations of $\mathcal{L}$ on the first 101 moments of each sequence give 1,128 strictly positive values. Complete outputs are in the files

`data/02-golden-moments-verification.json`
`data/02-golden-moments-verification.txt`

Some exact determinants are displayed below. Large integers have been computed directly, not inferred from the determinant identities.

n	$D_n^{(0)}(a)$	$D_n^{(1)}(a)$	$D_n^{(0)}(c)$	$D_n^{(1)}(c)$
1	1	1	1	1
2	2	4	1	2
3	20	100	4	20
4	1,440	20,864	100	1,440
5	970,624	45,897,152	20,864	970,624
6	7,618,255,104	1,300,185,348,608	45,897,152	7,618,255,104

The repetition is explained exactly by (15.4).

The next table shows the scaled consecutive ratios. These are rounded floating-point diagnostics obtained only after the integer coefficients were computed exactly. The theorem gives the limit 1; the table makes clear that the convergence is still slow at the displayed indices.

n	$\frac{a_{n+1}}{a_n}$	$\frac{\log n}{n}$	$\frac{c_{n+1}}{c_n}$	$\frac{\log n}{n}$
20	1.798430		1.743890	
50	1.568256		1.545876	
100	1.472240		1.461060	
200	1.410484		1.404905	
400	1.368842		1.366053	

No monotonicity of these *scaled* ratios is asserted from this table.

22.2 Enclosures far beyond the range of a power-series evaluation

Equation (12.8) has a useful exact numerical consequence. Choose $s_0 > 0$, set $s_{-1} = U(s_0)$, and define

$$s_{k+1} = s_k(1 + s_{k-1}) \quad (k \geq 0). \quad (22.1)$$

Then $U(s_k) = s_{k-1}$ for every $k \geq 0$. Indeed, $U^{-1}(s_k) = s_k(1 + U(s_k)) = s_k(1 + s_{k-1})$. Thus values of the true transform can be transported along its inverse orbit without any approximation to the functional equation.

For a rigorous seed, take $s_0 = 1/100$. Equation (16.4) gives

$$\sum_{j=0}^{31} (-1)^j a_j s_0^j \leq F(s_0) \leq \sum_{j=0}^{30} (-1)^j a_j s_0^j. \quad (22.2)$$

Both endpoints are rational and computed exactly. Positive additions and multiplications preserve lower and upper enclosures under (22.1). The script `code/02-golden-moments-orbit_bounds.py` propagates these enclosures with 100-digit Decimal arithmetic, rounding down for lower endpoints and up for upper endpoints. Correctly rounded logarithms, square roots and exponentials are enlarged by one adjacent representable value on each side before they are used in an enclosure [9]. Subtracted products are rounded in the opposite direction before subtraction.

For each row in the following table there is a true inverse-orbit point s_k in the full logarithmic interval recorded in `data/02-golden-moments-orbit_bounds.json`. The middle column is only a rounded location; the last column is a widened outward enclosure of

$$s_k^\alpha F(s_k) = \frac{s_{k-1}}{s_k^\beta}.$$

The enclosure concerns that true point, not every point in the interval.

k	approximate $\log s_k$	enclosure of $s_k^\alpha F(s_k)$
110	2.310822	[0.871023967670, 0.871023967671]
120	293.675802	[0.999996764030, 0.999996764031]
130	36119.735741	[0.999999973689, 0.999999973690]
140	4442433.820372	[0.999999999786, 0.999999999787]
150	546383240.170012	[0.999999999998, 0.999999999999]

These are finite computational enclosures conditional only on the exact identities proved above and the correctly rounded arithmetic operations. They illustrate the constant-one law at extremely large arguments. They are not its proof: Lemma 18.1 proves that law at all sufficiently large arguments, not merely along the selected orbit.

22.3 Reproduction and complexity

In the delivered package layout, where the three scripts and their data files carry no prefix, the checks and the delivered three-pass build were

```
python3 code/verify.py -n 401
python3 code/orbit_bounds.py
python3 code/make_tables.py
sh build.sh
```

The scripts write their outputs under those unprefixed names, not over the shipped files with the prefix 02-golden-moments-; the report's README gives the commands for rerunning them on a copy. The PDF of this report is built by its **Makefile**. The coefficient generator uses $N(N+1)(N+2)/6$ integer products and $O(N^2)$ integer storage cells. These are arithmetic-operation counts, not unit-cost bit-complexity claims: the coefficients grow rapidly. The small formal-series, Fock, and determinant checks use independent formulas to reduce the chance of an indexing or recurrence error. The PDF build requires a normal pdfLaTeX installation and the packages listed in the preamble; the mathematical checks do not require a computer algebra system.

23 Further research questions and concrete next steps

The moment representation changes the available methods: it allows orthogonal polynomials, resolvent bounds, probability tilting, and Tauberian arguments to be used alongside formal reversion. The following questions are not treated as established consequences of finite data. They are follow-up tasks for this construction, not claims that every item is a previously recognized open problem; existing general theorems may resolve some of them once their hypotheses are checked.

Research question 23.1 (Regularity and the support of the two laws). Are μ and ν absolutely continuous on $(0, \infty)$? Do their supports fill all of $[0, \infty)$, or can gaps or positive atoms occur?

The present proof gives unbounded support, no atom at zero, and precise mass near zero, but none of these excludes gaps or a singular component. A concrete first step is to check applicable free-convolution regularity theorems, and then continue the coupled Cauchy-transform identities into the upper half-plane and analyze boundary values. Any proposed density asymptotic must include a regularity argument; differentiating (12.10) is not sufficient.

Research question 23.2 (A full asymptotic equivalent for the coefficients). What is the multiplicative asymptotic equivalent of a_n ? In particular, does the finer Bell-number normalization proposed in Part I (Conjecture 8.1) converge to a finite positive constant?

Part I proposes a correction involving $\exp(W(n)^2 + 3W(n))$ relative to Bell numbers. The ratio law proved here is weaker: an error tending to zero in a single ratio can still accumulate into an unbounded logarithmic error. A practical next step is to derive a uniform expansion of the tilted-law mean beyond $n/\log n$, with a summable remainder after the proposed normalization.

Research question 23.3 (Second-order golden-ratio asymptotics). Can $F(s) = s^{-\alpha}(1 + o(1))$ be sharpened to an explicit second-order expansion, and does its correction exhibit oscillation under the inverse iteration $s_{k+1} = s_k(1 + s_{k-1})$?

The exact equation $d(x) = -\beta d(g(x)) - \beta \varepsilon(x)$ is a suitable starting point. Its negative contraction coefficient makes orbit-dependent alternating corrections plausible, but the present proof does not establish a global oscillatory profile or an optimal error rate.

Research question 23.4 (Effective convergence of the certified approximants). How quickly do $F_{2r}(s) - F_{2r+1}(s)$ tend to zero, uniformly when s varies over a growing interval? Can one obtain practical a priori stopping bounds for numerical evaluation?

The proof gives monotone convergence on fixed compact intervals, but not a useful global complexity estimate. The known finite-stage Fibonacci exponents indicate why the limits $r \rightarrow \infty$ and $s \rightarrow \infty$ require care. Interval evaluation with adaptive algebraic root isolation is a concrete computational approach.

Research question 23.5 (Orthogonal polynomials and continued fractions). What are the asymptotics of the Jacobi or Stieltjes continued-fraction coefficients associated with μ and ν ?

Strict Hankel positivity guarantees nondegeneracy of every finite orthogonalization step, while (15.4) ties the two systems together. Exact determinant tables can generate candidate recurrence coefficients, but the main challenge is to prove their large-degree behavior and connect it to the two endpoint laws.

Research question 23.6 (Sharper comparison of the companion sequence). Does $c_n/a_{n-1} \rightarrow 1$, and what is the first correction if it does?

Proposition 20.2 supplies only a polynomial-factor bound. The renewal recurrence suggests that endpoint terms should dominate its convolutions. Turning that observation into a uniform bound on the entire convolution, rather than on a fixed number of terms, would resolve the question without assuming the full asymptotics of a_n .

Research question 23.7 (Other reversion fixed points and positivity-preserving maps). Which variants of $M = 1 + zM^pB(zM^q)$ preserve normalized Stieltjes moment sequences, and which OEIS reversion sequences have fixed points of such maps?

The factorization in Section 13 proves the case $(p, q) = (2, 1)$. It is not a proof for arbitrary p, q . Positive block operators and noncrossing transforms offer testable sufficient mechanisms; negative Hankel minors provide exact obstructions in unsuccessful cases.

Research question 23.8 (A modular formalization in ProveIt). Can the proof be formalized with reusable lemmas for formal transforms, positive kernels, compact moment limits, and iteration rigidity?

A feasible first stage is the formal coefficient stabilization and the bivariate determinant identities, followed by the finite-remainder formula and the convex-slope lemma. The operator-to-measure limit is a separate analytic component. The scalar rigidity lemma is comparatively isolated and could be formalized independently. No code included here is presented as a completed Lean formalization.

24 Conclusion

A088714 and A088713 are not merely two rapidly growing coefficient lists. They are the determinate moment sequences of a coupled pair of positive measures. This representation proves every strict Hankel inequality, provides monotone algebraic bounds for the associated transforms, and makes their divergent ordinary generating functions meaningful on the negative axis through explicit Stieltjes and Borel–Laplace formulas.

The nonlinear inverse relation $U^{-1}(t) = t(1+U(t))$ forces complementary golden-ratio exponents, including their leading constants. Part I’s Bell-scale coefficient bound, together with the new convexity, determines consecutive ratios and the far-tail logarithmic law. The resulting picture is rigorous at both endpoints while leaving the finer coefficient normalization, measure regularity, and second-order behavior clearly open.

A Dependency and novelty audit

The main logical dependencies are as follows. The formal equation and the two compact positive-operator constructions prove existence of a limiting moment measure. The inherited growth bound supplies exponential integrability and determinacy, and also rules out bounded support. Moment positivity proves the strict Hankel and log-convexity statements. Weak convergence of compact approximants proves the real functional equation. Its inverse form, independently of any ratio asymptotics, proves the golden-ratio transform law. The self-contained Tauberian lemma then proves the small- t distribution-function laws. Separately, log-convexity plus the inherited growth bound proves the consecutive ratios, and these prove the far tails by size-biasing.

This separation avoids several invalid shortcuts: no divergent ordinary series is substituted at a positive real argument; no neighboring ratio is deduced just from an n th-root equivalent; no infinite zero or positivity statement is inferred from finite determinants; no cumulative mass asymptotic is differentiated to invent a density; and no numerical agreement is used to exchange the compact-approximation and large-argument limits.

The repository snapshot inspected for the source article (Part I) was commit

```
1f1981f682b2878bde51a6ad40c22777f362fc05
```

The selected prior TeX file has Git blob identifier

```
ccb50ef335ac4de27bd98a5a152fb48c978c7a96
```

and is located under

```
SetTheory/Cardinals/docs/reports/  
generating-functions-and-asymptotics/  
oeis-sequence-asymptotics/a088714-bell-scale-growth/
```

The paper’s own source and tests are independent deliverables, not changes to that repository. (Placement note, 2026-10-01: they now form Part II of that report, beside the unchanged Part I; see Section 11.) The accompanying source audit, shipped as `02-golden-moments-SOURCE_AUDIT.md`, records the scope of the literature check and the distinction between prior general machinery and the sequence-specific deductions made here.

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